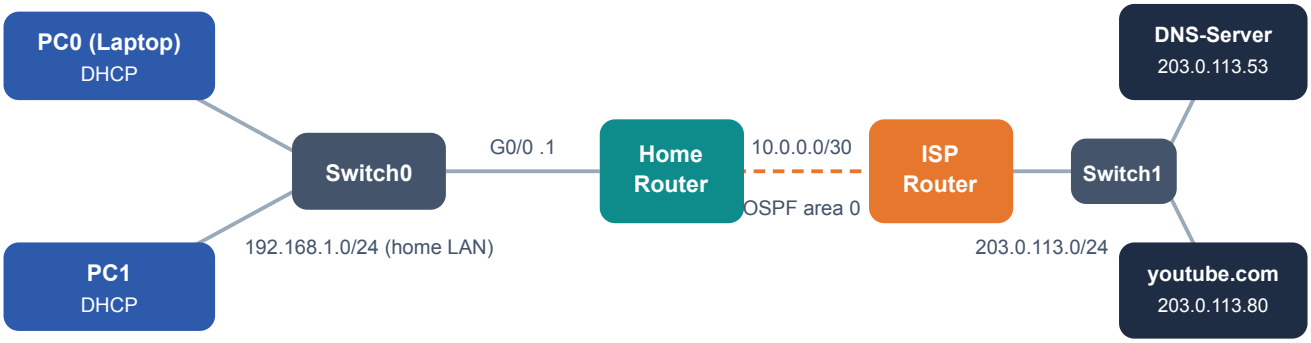
Packet Tracer live demo

Network MCP Server with a Diagnosis Agent · 21CSC302J · a small network you build in Cisco Packet Tracer, then break four times, once per layer, and diagnose live with the same checks our agent runs.

Ready-made file: `Network-Diagnosis-Demo.pkt` in this folder already has everything in section 3 built and tested: both routers configured with OSPF FULL, both PCs on DHCP, DNS answering `youtube.com`. Open it and go straight to section 4. Sections 2 and 3 are there if you want to rebuild it.

What the demo shows. In the project, the agent runs `ipconfig`, `ping`, `tracert` and `nslookup` for you and explains the result. In Packet Tracer we run the same commands by hand on a PC, in the same bottom-up order, so the class sees exactly how the agent reasons. About 6 minutes. Build it once beforehand (about 20 minutes) and save it as `Network-Diagnosis-Demo.pkt`.

1. Topology



2. Devices and addresses

Device (Packet Tracer model)	Port	Address	Notes
PC0, PC1 (PC-PT)	Fa0	DHCP from HomeRouter	Get 192.168.1.21 onward
Switch0 (2960-24TT)	Fa0/1, Fa0/2, G0/1	none	PC0 → Fa0/1, PC1 → Fa0/2, G0/1 → HomeRouter
HomeRouter (2911)	G0/0 G0/1	192.168.1.1/24 10.0.0.1/30	DHCP server for the LAN, OSPF
ISPRouter (2911)	G0/0 G0/1	10.0.0.2/30 203.0.113.1/24	OSPF
Switch1 (2960-24TT)	Fa0/1, Fa0/2, G0/1	none	Servers → Fa0/1, Fa0/2, G0/1 → ISPRouter
DNS-Server (Server-PT)	Fa0	203.0.113.53/24, gw .1	DNS service on
youtube.com (Server-PT)	Fa0	203.0.113.80/24, gw .1	HTTP service on (default)

Cables: **Copper Straight-Through** everywhere except HomeRouter G0/1 to ISPRouter G0/0, which is **Copper Cross-Over**.

3. Build it (about 20 minutes)

HomeRouter → CLI tab, paste:

```
enable
configure terminal
hostname HomeRouter
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
exit
ip dhcp excluded-address 192.168.1.1
192.168.1.20
ip dhcp pool HOME
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 203.0.113.53
exit
router ospf 1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
end
write memory
```

ISPRouter → CLI tab, paste:

```
enable
configure terminal
hostname ISPRouter
interface GigabitEthernet0/0
 ip address 10.0.0.2 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 ip address 203.0.113.1 255.255.255.0
 no shutdown
exit
router ospf 1
 network 10.0.0.0 0.0.0.3 area 0
 network 203.0.113.0 0.0.0.255 area 0
end
write memory
```

The same commands are in `HomeRouter-config.txt` and `ISPRouter-config.txt` beside this guide.

Servers (Desktop and Services tabs)

1. **DNS-Server:** Desktop → IP Configuration → Static: IP 203.0.113.53, mask 255.255.255.0, gateway 203.0.113.1, DNS 203.0.113.53. Services → DNS → **On**. Name `youtube.com`, Type A Record, Address 203.0.113.80 → Add.
2. **youtube.com server:** IP 203.0.113.80, mask 255.255.255.0, gateway 203.0.113.1. Services → HTTP stays On. Optional: edit `index.html` to say "YouTube (demo)".
3. **PC0 and PC1:** Desktop → IP Configuration → **DHCP**. They should get 192.168.1.21 and up, gateway 192.168.1.1, DNS 203.0.113.53.

Check it works before the demo

- HomeRouter: `show ip route` lists `O 203.0.113.0/24 via 10.0.0.2`. `show ip ospf neighbor` shows FULL.
- PC0 Command Prompt: `ping 203.0.113.80` replies, `nslookup youtube.com` gives 203.0.113.80, and Web Browser opens `http://youtube.com`.
- Switch to **Realtime** mode and wait until every link light is green. Save as `Network-Diagnosis-Demo.pkt`.

4. The live demo (about 6 minutes)

Keep PC0's Command Prompt open on the projector. For each scene: the presenter for that layer breaks it, PC0 runs the agent's checks from the bottom up, the class sees which check fails first, then it is fixed.

Scene 0 Everything working · Anbu

```
ipconfig                → 192.168.1.21, gateway 192.168.1.1, DNS 203.0.113.53
ping 192.168.1.1        → replies       (gateway)
ping 203.0.113.80       → replies       (server across the ISP)
tracert 203.0.113.80    → 192.168.1.1, 10.0.0.2, 203.0.113.80
nslookup youtube.com    → 203.0.113.80
```

Say: "These five commands are the agent's checklist, bottom to top. All pass, so the network is healthy. Now each of us will break one layer."

Scene 1 Physical layer: a dead link · Charith

Break: Switch0 → CLI: `enable`, `conf t`, `interface fa0/1`, `shutdown`. PC0's link light turns red.

```
ping 192.168.1.1        → Request timed out.   ✗ first check fails
```

Diagnosis: even the gateway next door does not answer and the link light is red, so it is the physical layer: cable, port or Wi-Fi. No need to test DNS.

Fix: `no shutdown` on fa0/1, wait for green, ping again.

Scene 2 Addressing: wrong gateway · Rupesh

Break: PC1 → IP Configuration → Static: IP 192.168.1.30, mask 255.255.255.0, gateway **192.168.1.100**. Run the checks on PC1.

```
ipconfig                → gateway 192.168.1.100
ping 192.168.1.1        → replies             (same LAN still works)
ping 203.0.113.80       → Request timed out.   ✗
```

Diagnosis: the LAN works but nothing leaves it. The gateway .100 is not our router (.1), so packets for other networks go nowhere.

Fix: set PC1 back to DHCP; the ping to 203.0.113.80 now replies.

Scene 3 Routing: the path breaks · Basith

Break: HomeRouter → CLI: `conf t`, `router ospf 1`, `no network 10.0.0.0 0.0.0.3 area 0`. Then on HomeRouter `show ip route`: the 203.0.113.0 route is gone.

```
ping 192.168.1.1          → replies
ping 203.0.113.80         → Destination host unreachable (from 192.168.1.1)  ✗
tracert 203.0.113.80      → 1  192.168.1.1   then unreachable
```

Diagnosis: traceroute stops right after the home router, and the router itself says it has no route. The problem is routing, not our PC.

Fix: `network 10.0.0.0 0.0.0.3 area 0` again. OSPF needs up to about 40 seconds to become FULL, so fill the gap by explaining hop-by-hop forwarding, then run `tracert` again.

Scene 4 Application: DNS down · Adhityan (the slide 17 story)

Break: DNS-Server → Services → DNS → **Off**.

```
ping 192.168.1.1          → replies
ping 203.0.113.80         → replies              (the Internet works)
nslookup youtube.com      → timed out             ✗
Web Browser http://youtube.com → Host Name Unresolved
```

Diagnosis: every lower layer passes, so the connection is fine; only the name lookup fails. This is exactly the agent's answer on slide 17: "your Internet works, but DNS is not answering". Show that `http://203.0.113.80` still opens the page.

Fix: DNS → On, then `nslookup youtube.com` and the browser work again.

Before you present: rehearse once on the lab PC, open the saved file, wait for green lights, run Scene 0 once before the class arrives (the very first ping and `nslookup` after opening can time out once while the devices learn each other's MAC addresses), and keep the four "break" steps on a sticky note. If the OSPF fix in Scene 3 is slow, keep talking; it recovers on its own.